

Deggendorf Institute of Technology

Master's Thesis

A Demonstration Thesis Using LaTeX

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A Demonstration Thesis Using LaTeX

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Declaration of Authorship

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Abstract

A fictional study of a classroom sand timer illustrates an engineering thesis. The teaching problem is how to connect a measurable requirement to a design choice and a transparent verification result. The objective is to compare two imaginary timer concepts against a target duration of sixty seconds. A simple duration model and three invented observations per concept demonstrate the method. Concept A has a mean duration of sixty seconds, while concept B has a mean duration of sixty-five seconds. The first concept meets the illustrative mean-duration requirement. The small constructed dataset cannot support claims about real devices or a wider population. Further work would require a physical prototype, calibrated measurements and a justified sampling plan. The example shows how sources, equations, figures and results can form a traceable argument. All observations are fictional and exist only to teach document structure.

Keywords: demonstration; requirements; verification; LaTeX

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The fictional author thanks the imaginary reading group for testing the clarity of this educational example.

Contents

Declaration of Authorship	I
Abstract	III
Acknowledgements	V
List of Figures	IX
List of Tables	XI
List of Code Listings	XIII
List of Abbreviations	XV
List of Symbols	XVII
1 Introduction	1
1.1 Background and objective	1
1.1.1 Illustrative hypothesis	1
1.2 Structure of the thesis	1
2 State of the Art	3
2.1 Background and source quality	3
2.2 Gap and comparison	3
3 Requirement Management	5
3.1 Measurable acceptance criterion	5
3.2 Traceability	5
4 Development of Concepts	7

4.1 Alternatives and selection	7
4.2 Development workflow	7
5 Development of Product	9
5.1 Design definition	9
5.2 Risk and reproducibility	9
6 Simulation	11
6.1 Model and analysis method	11
6.2 Fictional code example	11
6.3 Simulation boundary	11
7 Realization of Prototype	13
7.1 Prototype record	13
7.2 Measurement plan	13
8 Verification of Prototype	15
8.1 Illustrative results	15
8.2 Verification and limitations	15
9 Conclusion and Outlook	17
9.1 Conclusion	17
9.2 Outlook	17
References	19
Appendix	21
A Additional Example Material	23
A.1 Data and file formats	23
A.2 AI usage record format	23

List of Figures

4.1 Requirements lead to design and verification in the fictional study. Original teaching diagram. 7

List of Tables

8.1 Fictional durations for two imaginary concepts; every entry is constructed for teaching. 15

List of Code Listings

6.1	Fictional Python example that calculates an arithmetic mean.	11
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List of Abbreviations

API Application Programming Interface

PDF Portable Document Format

List of Symbols

Symbol	Description	Unit
i	Observation index	n.u.
n	Number of observations	n.u.
$\bar{t}$	Arithmetic mean duration	s
t_i	Duration of observation i	s

1 Introduction

1.1 Background and objective

This fictional study concerns a classroom sand timer. Its purpose is to teach how an engineering argument connects a requirement, a design and a verification. No real product is investigated.

1.1.1 Illustrative hypothesis

The illustrative hypothesis is that concept A gives a smaller mean duration error than concept B under identical conditions. This is a teaching proposition, not a claim of research novelty. A real thesis must identify an unresolved, measurable research hypothesis beyond the existing state of the art.

1.2 Structure of the thesis

Chapter 2 establishes the background. Requirements guide concept selection and development. Simulation precedes prototype realization. Chapter 8 presents the constructed results; the conclusion addresses their limitations and next steps.

2 State of the Art

2.1 Background and source quality

The references here teach citation mechanics rather than timer research. Knuth [1] provides a book example about typesetting. The journal article by Shannon is a separate example [2]. A public web publication supplies the online-reference example [3].

Multiple sources can be cited together [1], [2]. These references do not establish that either imaginary timer concept is superior.

2.2 Gap and comparison

A real literature study would compare timing mechanisms, measurement uncertainty and operating conditions. The gap must justify the hypothesis in Section 1.1.1. The present example makes no novelty claim.

3 Requirement Management

3.1 Measurable acceptance criterion

Requirement R1 states that the mean of the three illustrative durations shall be within 2 s of a target of 60 s. The criterion is evaluated in Chapter 8. For an actual study, a deadline, operating conditions and a justified sample size would be agreed before data collection.

3.2 Traceability

The mean is computed using Equation 6.1. Table 8.1 provides the observations and Appendix A identifies their source. This link makes a requirement testable rather than merely aspirational.

4 Development of Concepts

4.1 Alternatives and selection

Concept A uses one imaginary container shape; concept B uses another. The teaching example prioritizes mean-duration error and ease of assembly. A real comparison would justify the criteria, their weights and the evidence behind every score. No weighted score is invented here to imply real superiority.

4.2 Development workflow

Figure 4.1 shows the sequence used to organise this example. Requirements guide design; verification returns evidence about the initial target.

Requirements —————> Design —————> Verification

Figure 4.1: Requirements lead to design and verification in the fictional study. Original teaching diagram.

5 Development of Product

5.1 Design definition

The fictional product consists of two containers and a connecting opening. A real design chapter would specify dimensions, materials, interfaces and assembly tolerances. Here those details are deliberately absent because no device was built.

5.2 Risk and reproducibility

An illustrative failure mode is leakage at the joint. A real development process would assess its consequences, likelihood, detection and mitigation before testing. Versioned drawings and an assembly procedure would support reproducibility. The model in Chapter 6 illustrates the analysis step.

6 Simulation

6.1 Model and analysis method

The demonstration reduces each set of durations to its arithmetic mean. Equation 6.1 adds the durations and divides the total by their count.

$$\bar{t} = \frac{1}{n} \sum_{i=1}^n t_i. \quad (6.1)$$

The observation t_i and mean $\bar{t}$ are measured in seconds. The count n and index i have no unit. With $n = 3$, the sum for concept A is 180 s; division by three gives 60 s.

6.2 Fictional code example

Listing 6.1 implements only the arithmetic mean used in this teaching example. It is independently written demonstration code and is unrelated to any real thesis, company system or proprietary implementation.

```
1 """Fictional teaching example; unrelated to any real thesis or product."""
2
3
4 def arithmetic_mean(values):
5     """Return the arithmetic mean of a non-empty sequence."""
6     if not values:
7         raise ValueError("At least one value is required")
8     return sum(values) / len(values)
9
10
11 fictional_durations = [59, 60, 61]
12 print(arithmetic_mean(fictional_durations))
```

Listing 6.1: Fictional Python example that calculates an arithmetic mean.

6.3 Simulation boundary

This arithmetic example is not a physical sand-flow simulation. A real simulation chapter must define governing equations, boundary conditions, solver settings, con-

vergence checks and comparison with independent evidence.

7 Realization of Prototype

7.1 Prototype record

No physical prototype was built for this demonstration. In a real thesis this section would identify the realized configuration and deviations from the design. An assembly record would state component versions and relevant instruments.

7.2 Measurement plan

The fictional plan uses three observations per concept under identical assumed conditions. A real protocol would define the timing event, operator procedure, calibration, environmental conditions and measurement uncertainty. Invented data must never be relabelled as measurements. The demonstration data are kept in a machine-readable file as well as in Table 8.1.

8 Verification of Prototype

8.1 Illustrative results

Table 8.1 shows all six constructed durations. Concept A has a mean of 60 s; concept B has a mean of 65 s. The observations are fictional, not experimental evidence.

Concept	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Mean (s)
A	59	60	61	60
B	64	65	66	65

Table 8.1: Fictional durations for two imaginary concepts; every entry is constructed for teaching.

8.2 Verification and limitations

Using Equation 6.1, concept A meets R1 in Section 3.1, whereas concept B does not. Only the illustrative mean-duration criterion is assessed. The tiny invented sample cannot quantify reliability or support inference about real timers. A real evaluation would report uncertainty, repeatability and potential bias.

9 Conclusion and Outlook

9.1 Conclusion

The fictional numbers give concept A a smaller mean-duration error than concept B. The calculation in Section 8.1 addresses the teaching hypothesis from Section 1.1.1. It does not demonstrate a physical effect. The requirement-to-result chain is complete only within this constructed example.

9.2 Outlook

Next steps for a real study would include building the selected design, calibrating the timing method and justifying a larger sampling plan. The influence of operating conditions remains unanswered. These actions would turn a document demonstration into an evidence-producing investigation.

References

- [1] D. E. Knuth, *The TeXbook*. Reading, MA: Addison-Wesley, 1984.
- [2] C. E. Shannon, "A mathematical theory of communication," *The Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948. DOI: 10.1002/j.1538-7305.1948.tb01338.x.
- [3] T. Berners-Lee. "Re: Qualifiers on hypertext links...," Accessed: Sep. 7, 2026. [Online]. Available: <https://www.w3.org/People/Berners-Lee/1991/08/art-6484.txt>.

Appendix

Supporting material is separated from the main argument. The following attachment contains the small, fictional dataset used in this educational example.

A Additional Example Material

A.1 Data and file formats

The file `04_raw_data/example_durations.csv` contains the invented data. CSV means comma-separated values: a text format that spreadsheet tools can read. The compiled document uses Portable Document Format (PDF). An Application Programming Interface (API) is another acronym example; no external service is used by this project.

A.2 AI usage record format

For actual permitted AI assistance, record tool name, exact model, date, purpose, verbatim prompt and any supplied file or public source. Preserve prompts as text, not screenshots. Do not upload confidential or personal material.

Fictional format example: Tool: [tool]; Model: [model]; Date: [date]; Purpose: spelling review; Prompt: "Identify spelling errors in this sentence."; Supplied files: none. This is a format example, not an actual usage log.